

Koa Shen

B.S. Candidate in Mechanical Engineering · UC Santa Barbara

koashensf@gmail.com (415) 810-8344 koashen.com San Francisco Bay Area / Santa Barbara, CA

October 2, 2026

Hiring Team

Parker Hannifin

Re: Summer 2027 Quality Engineering Internship

Dear Parker Hannifin Hiring Team,

I am a mechanical engineering student at UC Santa Barbara applying for a Summer 2027 quality engineering internship. I am interested in Parker Hannifin because its motion and control technologies operate in demanding applications where product quality depends on sound design, consistent manufacturing, and disciplined verification. I would bring hands-on experience translating requirements into controlled designs, checking critical features, and using test results to reduce risk.

At SLAC National Accelerator Laboratory, I built an open-source Python and Drake digital-twin framework for tightly packaged motion assemblies. Its live collision checks help engineers identify interference risks and inform design changes before hardware is operated. I also produce PDM-controlled design documentation in Siemens Solid Edge, applying GD&T for manufacture and inspection, and perform structural checks on stages and mounts protecting detectors valued at more than \$100,000. This work has taught me to connect design intent, inspection requirements, and failure prevention.

With UCSB Gaucho Racing, I CNC-machine 4130 steel and 7075-T6 aluminum suspension components and work with the design team to account for GD&T and manufacturability. I held critical clevis and rod-end bores within a 0.0004-inch tolerance window and bearing interfaces within 0.0005 inches, verifying critical fits with go/no-go tests. This experience has shown me how clear requirements, process choices, and inspection methods work together to achieve conforming parts. In a personal robotics project, I also bench-validated encoder feedback that detects missed steps and corrects position error.

I would bring Parker Hannifin a practical quality mindset grounded in precision manufacturing, inspection, design documentation, and engineering verification. I would be excited to help improve product reliability and manufacturing quality while learning from a team responsible for critical motion and control systems.

Thank you for your consideration.

Sincerely,

Koa Shen